

Quick Activity 2: Canteen Group Order

A canteen sells rice meals at P85 each and drinks at P25 each. Every order receives a 10% discount on the rice-meal subtotal only. The canteen also charges a P5 packaging fee per rice meal. Develop an algorithm that calculates the final bill and the customer's change.

Rules: Drinks and packaging fees receive no discount. Add the packaging fee after calculating the discount. Quantities are positive whole numbers and payment covers the bill.

INPUT	PROCESS	OUTPUT
• riceMeals • drinks • payment	$\text{mealSubtotal} \leftarrow \text{riceMeals} \times 85$ $\text{drinkSubtotal} \leftarrow \text{drinks} \times 25$ $\text{discount} \leftarrow \text{mealSubtotal} \times 0.10$ $\text{discountedMealSubtotal} \leftarrow \text{mealSubtotal} - \text{discount}$ $\text{packagingFee} \leftarrow \text{riceMeals} \times 5$ $\text{finalBill} \leftarrow \text{discountedMealSubtotal} + \text{drinkSubtotal} + \text{packagingFee}$ $\text{change} \leftarrow \text{payment} - \text{finalBill}$	• finalBill • change

DECOMPOSITION:

1. Read riceMeals, drinks, and payment amount.
2. Calculate $\text{mealSubtotal} = \text{riceMeals} \times 85$.
3. Calculate $\text{drinkSubtotal} = \text{drinks} \times 25$.
4. Calculate $\text{discount} = \text{mealSubtotal} \times 0.10$.
5. Calculate $\text{discountedMealSubtotal} = \text{mealSubtotal} - \text{discount}$.
6. Calculate $\text{packagingFee} = \text{riceMeals} \times 5$.
7. Calculate $\text{finalBill} = \text{discountedMealSubtotal} + \text{drinkSubtotal} + \text{packagingFee}$.
8. Calculate $\text{change} = \text{payment} - \text{finalBill}$.
9. Display finalBill and change to two decimal places.

PSEUDOCODE:

```
START
  READ riceMeals, drinks, payment
  mealSubtotal  $\leftarrow$  riceMeals  $\times$  85
  drinkSubtotal  $\leftarrow$  drinks  $\times$  25
  discount  $\leftarrow$  mealSubtotal  $\times$  0.10
  discountedMealSubtotal  $\leftarrow$  mealSubtotal - discount
  packagingFee  $\leftarrow$  riceMeals  $\times$  5
  finalBill  $\leftarrow$  discountedMealSubtotal + drinkSubtotal + packagingFee
  change  $\leftarrow$  payment - finalBill
  DISPLAY finalBill, change
END
```

DRY-RUN CALCULATION: QUICK ACTIVITY TWO

CASE TEST ONE:

3 Rice Meals, 4 Drinks, Payment = P500

STEP	VARIABLE	CALCULATION	RESULT
1	mealSubtotal	$3 \times \text{P}85.00$	P255.00
2	drinkSubtotal	$4 \times \text{P}25.00$	P100.00
3	discount	$\text{P}255.00 \times 0.10$	P25.50
4	discountedMealSubtotal	$\text{P}255.00 - \text{P}25.50$	P229.50
5	packagingFee	$3 \times \text{P}5.00$	P15.00
6	finalBill	$\text{P}229.50 + \text{P}100.00 + \text{P}15.00$	P344.50
7	change	$\text{P}500.00 - \text{P}344.50$	P155.50

EXPECTED OUTPUT:

Final bill: P344.50

Change: P155.50

CASE TEST TWO:

5 Rice Meals, 3 Drinks, Payment = P600

STEP	VARIABLE	CALCULATION	RESULT
1	mealSubtotal	$5 \times P85.00$	P425.00
2	drinkSubtotal	$3 \times P25.00$	P75.00
3	discount	$P425.00 \times 0.10$	P42.50
4	discountedMealSubtotal	$P425.00 - P42.50$	P382.50
5	packagingFee	$5 \times P5.00$	P25.00
6	finalBill	$P382.50 + P75.00 + P25.00$	P482.50
7	change	$P600.00 - P482.50$	P117.50

EXPECTED OUTPUT:

Final bill: P482.50

Change: P117.50

EXPLANATION:

Applying the 10% discount to the entire bill would incorrectly reduce the price of drinks and packaging fees, which are explicitly non-discountable. It violates the rules by giving an unauthorized discount on items that must be charged at full price.